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Differential controls of warming and phosphorus enrichment on CO_2, CH_4, and N_2O emissions in shallow lake mesocosms☆

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ABSTRACT

Climate warming and eutrophication are reshaping greenhouse gas (GHG) emissions from shallow lakes, yet how different warming regimes interact with phosphorus enrichment remains poorly understood. Here, we conducted a year-long experiment using shallow-lake mesocosms planted with aquatic macrophytes to compare constant warming (a sustained 4°C increase) with variable warming (the same mean warming combined with short-term temperature fluctuations), together with phosphorus enrichment. Responses differed markedly among gases. Warming reduced annual CO_2 emissions, primarily through indirect changes in macrophyte communities and water chemistry. In contrast, warming increased CH_4 emissions mainly through direct thermal effects, although the increase was smaller under variable than under constant warming, reflecting distinct shifts in macrophyte functional composition under the two warming regimes. N_2O fluxes were largely insensitive to warming but responded primarily to phosphorus enrichment, which shifted the system from a net annual source to a sink. These findings demonstrate that greenhouse-gas responses to global change cannot be generalized across gases and highlight the need to account for warming regime, nutrient status, and macrophyte functional composition when predicting climate feedbacks from shallow lakes.

1. Introduction

Emissions of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) from inland waters have received increasing attention in recent decades (Lauerwald et al., 2023; Bhushan et al., 2024). Inland waters, encompassing rivers, lakes, reservoirs, and wetlands, cover approximately 3.6×10^6 km^2, accounting for approximately 2.67% of the global land surface (Feng et al., 2016; Gao et al., 2025). Lakes represent the dominant component, comprising more than 90% of inland-water area (Raymond et al., 2013). As active regulators of carbon and nitrogen biogeochemical cycling, lakes are increasingly recognized as important hotspots of GHG emissions to the atmosphere (Hughes-Allen et al., 2021; Leng et al., 2023). Shallow lakes, owing to their small volume and limited mean depth, respond sensitively to warming and to changes in mixing regimes and they contribute disproportionately to inland-water fluxes of CO_2, CH_4, and N_2O relative to their surface area (Bastviken et al., 2011; Holgerson and Raymond, 2016; Jankowski et al., 2006). Consequently, understanding how climate forcing alters GHG emissions from shallow lakes is critical for constraining regional and global budgets (Davidson et al., 2015; Downing et al., 2006; Raymond et al., 2013).

Previous work has shown that warming can strengthen thermal stratification, promote near-bottom oxygen depletion, and shift primary producer composition, thereby reshaping the pathways of GHG production, consumption, and transport (Jöhnk et al., 2008; Stetler et al.,

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2021; Yvon-Durocher et al., 2010a). Warming has been shown to significantly regulate CO_2 and CH_4 emissions from shallow lakes through combined effects on ecosystem metabolism and physical gas exchange (Marotta et al., 2014; Yuan et al., 2024; Yvon-Durocher et al., 2010b). Beyond CO_2 and CH_4, N_2O fluxes are tightly regulated by temperature, as both nitrification and denitrification are enzyme-mediated processes with strong thermal sensitivity (Velthuis and Veraart, 2022; Veraart et al., 2011). However, most existing studies have emphasized long-term or gradual warming, whereas the biogeochemical consequences of variable warming events remain poorly constrained (Xiao et al., 2022). Variable warming can transiently alter redox boundaries and microbial activity, thereby modulating N_2O yields once mixing resumes (Voigt and Shaw, 2015; Yvon-Durocher et al., 2014). Increasing evidence further suggests that lake heatwaves not only modify gas transport pathways but also intensify internal carbon cycling by accelerating organic-matter mineralization and methanogenesis, while constraining aerobic oxidation under prolonged oxygen limitation, ultimately amplifying net CO_2 and CH_4 release (Reichstein et al., 2013; Zachos et al., 2008). Therefore, a comprehensive assessment of both sustained warming and pulsed, variable warming events is essential. Such an assessment can help clarify the mechanisms through which temperature regulates GHG emissions in shallow lakes and improves our understanding of their pronounced seasonal variability.

In addition to climate warming, eutrophication represents a major and pervasive stressor on lake ecosystems, and in some cases may outweigh warming in regulating GHG emissions (Davidson et al., 2015; Jeppesen et al., 2010). Lake CO_2 dynamics are closely tied to metabolic balance, which is jointly modulated by nutrient enrichment, such that eutrophication can substantially shift CO_2 source-sink behavior (Grasset et al., 2020; Scharfenberger et al., 2019; Xiao et al., 2024). Additionally, nutrient enrichment promotes phytoplankton blooms and organic matter accumulation, leading to reduced light penetration and increased sediment oxygen demand. These changes favor the development of low-oxygen conditions and thereby alter the pathways governing CH_4 production and emission (Beaulieu et al., 2019; Davidson et al., 2018; Kosten et al., 2012). For N_2O, nutrient enrichment regulates nitrification–denitrification dynamics (Seitzinger, 1988; Velthuis and Veraart, 2022; Veraart et al., 2011).

Among nutrients, phosphorus has gained increasing attention as a key driver of freshwater eutrophication, as it commonly acts as the primary limiting nutrient in lakes and exerts strong control over plankton biomass, community structure, and microbial activity (Schindler, 1977; Liang et al., 2020; Xu et al., 2025). By stimulating primary production and altering the balance between photosynthesis and respiration, phosphorus enrichment can shift lakes toward net heterotrophy or induce pronounced seasonal reversals in CO_2 fluxes (Scharfenberger et al., 2019). In parallel, phosphorus-driven increases in organic matter supply and changes in microbial assemblages can enhance anaerobic metabolic pathways, thereby indirectly promoting CH_4 production and modulating coupled carbon–nitrogen cycling processes that influence N_2O emissions (Davidson et al., 2015; Fiskal et al., 2019). Growing evidence suggests that phosphorus may play a more important role than nitrogen in controlling CO_2 production in eutrophic shallow lakes (Yuan et al., 2024). Thus, accurately assessing greenhouse gas emissions from lake waters requires an integrated consideration of phosphorus enrichment and temperature variation.

Based on these considerations, shallow-lake mesocosms were employed within a unified experimental framework that explicitly integrates multiple biotic and abiotic drivers to assess the individual and interactive effects of constant warming, variable arming, and phosphorus enrichment on GHGs emissions and their underlying mechanisms. The study addresses three key questions: (1) whether CO_2, CH_4, and N_2O exhibit differential responses to warming (including constant and variable warming) and phosphorus enrichment; (2) how interactions between warming and phosphorus enrichment modify the production and emission characteristics of these gases through coupled biological and physicochemical processes; and (3) which key biogeochemical pathways regulate variations in CO_2, CH_4, and N_2O emissions under warming and eutrophication. This framework advances a process-based understanding of greenhouse gas dynamics in shallow lakes and their responses to ongoing environmental change.

2. Materials and methods

2.1. Experiment set-up

A comprehensive outdoor experimental mesocosm system consisting of thirty-six insulated cylindrical polyethylene mesocosms (volume = 2500 L, diameter = 1.5 m, height = 1.4 m) is situated at Huazhong Agricultural University in Wuhan (30°29′N; 114°22′E), China. Each mesocosm supplied with 600 W heating devices (A10-3, Xin Shao Guang, China) and temperature sensors (DS18B20, Maxim IC, USA) (Fig. S1). In October 2017, surface sediment was collected from the open water area of Liangzi Lake (30°11′3″N; 114°37′59″E) and thoroughly mixed in a clean container. Liangzi Lake is a mesotrophic lake (TLI 42.5) (Xu et al., 2020); with the water total nitrogen (TN) and total phosphorus (TP) concentrations in our sampling area of about 0.432 mg L^{-1} and 0.023 mg L^{-1}, respectively (Li et al., 2018). Subsequently, a 10 cm-deep sediment layer was laid at the bottom of each mesocosm, followed by the injection of water collected from Liangzi Lake through a 20 μm mesh filter to a depth of 1 m. The sediment of the lake contains propagules of submerged plants and floating-leaf plants, which can naturally emerge. Before the start of the experiment, all the mesocosms were left for colonization at ambient conditions for two months (November–December 2017).

We applied a fully factorial design comprising two stressors treatments with the following levels: (1) warming with three levels, including ambient control, constant warming, and variable warming. The constant warming treatment was maintained at 4°C above ambient control conditions throughout the experiment, mimicking the projected temperature increase under RCP 8.5 in the region by the end of the century (Chen and Frauenfeld, 2014; Pachauri et al., 2014). The variable warming treatment was designed to represent increased temperature variability rather than a further increase in mean temperature. It therefore shared the same 4°C baseline increase as the constant warming treatment, but 17 discrete temperature fluctuation events were superimposed on this baseline during the experimental year. These events occurred intermittently throughout the experiment, lasted 3–12 days each, and had an overall intensity ranging from 0 to 8°C, with deviations occurring both above and below the 4°C warming baseline to represent heat and cold events, respectively (Guo et al., 2017; Pachauri et al., 2014). Thus, the variable warming treatment differed from constant warming mainly in the timing, frequency, direction, duration, and amplitude of temporary temperature fluctuations, while retaining the same baseline warming level. (2) Phosphorous enrichment with two levels including a control no phosphorus addition, and phosphorus enrichment involving phosphorus loading. Phosphorous enrichment concentrations and frequency were set based on previous regional studies of the phosphorus budget (input minus output) to lakes in the middle and lower reaches of the Yangtze River adequately adapted to the temporal scale of our experiment (Gao et al., 2016). The experiment involved a total of six treatments, each replicated six times and randomly allocated to thirty-six mesocosms.

The constant warming and variable warming treatments were achieved by a computer-controlled temperature system (Xu et al., 2020). The dose of phosphorus addition was determined by referring to previous studies (Gao et al., 2016; Huang and Xu, 2001), and comprising the addition of 25 μg L^{-1} phosphate (KH_2PO_4) to the corresponding mesocosms every two weeks to mimic the eutrophication process. The experiment ran for a total of 364 days between January 1 and December 31, 2018, with the heating treatment starting on the first day and the addition of phosphorus beginning in January 17. Water levels in the

mesocosms were maintained constant by replenishing tap water to offset evaporation losses throughout the experiment, except when compensated for by rainfall. Although there were occasional days of precipitation in February and March 2018, the rise in water levels due to rainfall never exceeded 10 cm above the initial conditions.

2.2. Sampling and processing

2.2.1. Environmental factors measurements

Throughout the entire experiment, water samples for the collection and analysis of physical, chemical factors, and biological factors were collected and analyzed in the mesocosms. Sampling occurred biweekly in the first two months (January and February) and weekly in the subsequent months (March to December). Sampling in the variable-warming treatment followed this regular monitoring schedule and was not intentionally synchronized with the onset or termination of individual temperature-fluctuation events. Therefore, the measured environmental variables reflect the integrated effects of the variable-warming regime over the preceding period rather than only the immediate short-term response after a single pulse.

Water temperature (WT) was continuously monitored using temperature sensors integrated into the computer-controlled warming system. Dissolved oxygen (DO), pH and electrical conductivity (EC) were measured in situ using a portable multiparameter meter (HQ60d, HACH, USA). Turbidity (Turb) was measured by a portable turbidity meter (WGZ-2B, Xinrui, China). Integrated water-column samples were collected from each mesocosm using a Plexiglas tube sampler (70 mm inner diameter, 1 m length). After gentle homogenization, each sample was divided into subsamples for analyses of nutrients, chlorophyll *a* (Chl-*a*) and dissolved organic carbon (DOC). Concentrations of total nitrogen (TN), total phosphorus (TP), phosphate (PO_4^{3-}), nitrate (NO_3^-), and ammonium (NH_4^+) were determined spectrophotometrically (Cleverchem380, DeChem-Tech., Germany) according to the standard methods for water and wastewater analysis (The State Environmental Protection Administration the Water and Wastewater Monitoring Analysis Method Editorial Board, 2002). For Chl-*a* analysis, 1000 mL of water was filtered through a 1.2 μm Whatman GF/C glass-fiber filter. The retained pigments were extracted with ethanol, and Chl-*a* concentrations were determined spectrophotometrically using a UV-2800 spectrophotometer (Unico, China). For DOC analysis, 50 mL of water was filtered through a 0.22 μm mixed cellulose ester (MCE) membrane filter (Millipore, USA), and the filtrate was analyzed monthly using a Vario TOC analyzer (Elementar, Germany). Macrophyte community structure was assessed separately for floating-leaved and submerged macrophytes. Floating-leaved macrophyte coverage (FPC) was visually estimated as the percentage of the water surface occupied by floating-leaved vegetation. The percent volume inhabited (PVI) of submerged macrophytes was calculated as the product of submerged plant coverage and mean plant height divided by water depth. FPC and PVI were used as proxies for the relative abundance of floating-leaved and submerged macrophytes, respectively.

2.2.2. GHG measurements and flux calculations

GHGs fluxes were determined using the floating static chamber method. The floating static chamber consists of a polyethylene cylindrical container with one end open (inner diameter = 45 cm, total height = 50 cm). The outer surface of the chamber is equipped with uniformly distributed buoyant rings, and the chamber is fixed at a certain height to ensure that the lower part of the chamber is submerged approximately 5 cm below the water surface during sampling, thereby maintaining the chamber's airtightness. The operational components of the floating chamber include pressure balance tubes, sampling tubes, and temperature probes. The exterior of the chamber is entirely covered with aluminum foil to reflect sunlight and minimize internal heating. During the experiment, on a clear day in the middle and late of each month, gas sampling was conducted from 8:30 to 11:30. As with the environmental sampling, monthly gas sampling in the variable-warming treatment followed the fixed sampling schedule and was not deliberately conducted immediately after each temperature-fluctuation event. Before each batch of gas collection, the floating chamber was allowed to rest horizontally for 3 min to ensure thorough mixing with the surrounding air. Subsequently, the chamber was gently placed on the water surface and left undisturbed until the water surface was calm. Gas samples were then extracted at 2 min, 14 min, and 26 min, totaling 3 times. During the first sampling, the syringe was emptied twice before filling it with 30 mL of gas. For the subsequent two samplings, the syringe was emptied only once before extraction. Following the completion of gas sample collection, GHGs were measured within 48 h.

The GHGs samples were injected into the CA-6 manual gas injection system using a 30 mL medical syringe equipped with a three-way valve, and then analyzed using the GC 7890 gas chromatography (Agilent, USA). The column oven temperature was uniformly set to 55°C. The CO_2 concentration was determined using a flame ionization detector (FID) with a working temperature of 200°C. The system included a 2 mm inner diameter, 2 m long 60–80 mesh Porapak Q stainless steel separation column and a nickel catalyst converter with a working temperature of 375°C. High-purity N_2 (>99.999% purity) was used as the carrier gas at a flow rate of 25 $cm^3\ min^{-1}$, while the flow rates of air and high-purity H_2 were set at 400 $cm^3\ min^{-1}$ and 30 $cm^3\ min^{-1}$, respectively. The CH_4 concentration was also measured using an FID with a working temperature of 200°C. The system mainly comprised a 2 mm inner diameter, 2 m long 60–80 mesh 13XMS stainless steel separation column. High-purity N_2 was used as the carrier gas at a flow rate of 30 $cm^3\ min^{-1}$, with the flow rates of air and high-purity H_2 set at 400 $cm^3\ min^{-1}$ and 30 $cm^3\ min^{-1}$, respectively. The N_2O concentration was determined using an electron capture detector (ECD) with a working temperature of 300°C. Separation and backflushing of N_2O were achieved using two 80–100 mesh Porapak Q stainless steel separation columns, one with a length of 1 m and the other with a length of 3 m. High-purity N_2 was used as the carrier gas at a flow rate of 25 $cm^3\ min^{-1}$.

The rate of change of gas concentration over time (slope of the regression curve, to ensure a good linear relationship, only experimental data with $R^2 \geq 0.90$ were selected) can be obtained through calculation. The fluxes equation is as:

$$F = \frac{M}{V_0} \times \frac{P}{P_0} \times \frac{T_0}{T} \times \frac{V}{A} \times \frac{dc}{dt} \quad (1)$$

where F is the GHG flux (mg $m^{-2}h^{-1}$); M is the molar mass of CO_2, CH_4, or N_2O (44, 16, and 44 g mol^{-1}, respectively); P_0 is standard atmospheric pressure (101.325 kPa); T_0 is standard absolute temperature (273.15 K); V_0 is the molar volume of an ideal gas at standard conditions (22.4 L mol^{-1}); V/A represents the effective chamber height above the water surface (m), where V is the chamber headspace volume (m^3) and A is the covered water-surface area (m^2); P and T are the actual pressure (kPa) and absolute temperature (K) inside the chamber during sampling; and dc/dt is the slope of CO_2, CH_4, or N_2O concentration change inside the chamber (ppm h^{-1}).

By performing addition calculations on the monthly average CO_2, CH_4, or N_2O fluxes data within each treatment group during the experiment, the total annual flux for CO_2, CH_4, or N_2O can be respectively represented. If the total annual flux is negative, the treatment group acts as an atmospheric sink over the course of the year, whereas if the total annual flux is positive, the treatment group acts as an atmospheric source over the course of the year. The total annual flux equation is as:

$$CE = \sum_{i}^{n} F_i \times t_i \quad (2)$$

where *CE* (Cumulative exchange) is the total annual fluxes of CO_2, CH_4, or N_2O (g $m^{-2}\ y^{-1}$), *n* is the total number of sampling times throughout the year, *Fi* is the average CO_2, CH_4, or N_2O fluxes for the *i* month, and t_i

is the number of days in the *i* month.

2.3. Statistical analysis

All statistical analyses were performed with R software version 4.4.3 (R Core Team, 2024). Due to only experimental data with $R^2 \geq 0.90$ being selected when calculating the rate of change in gas concentration over time, some values were missing in the gas fluxes data. We statistically imputed these missing values with a machine learning algorithm called "missForest". This method uses a random forest trained on the observed values of a data matrix to predict the missing values and automatically calibrates the imputed values through a set of iterations (Fig. S2). Multiple generalized linear mixed-effect models (lmer function in lme4 package), fitting normal distributions, were applied to analyze effects of temperature (ambient temperature, constant warming, variable warming), phosphorus treatments (control and phosphorus loading), and their interactions on environmental factors and GHG fluxes during the experiments, with sampling month as random factor. Sampling month was included to account for shared temporal variation and seasonal dependence among repeated observations. For non-normally distributed data, natural logarithm or square root transformation was applied. Statistical significances were assessed using type III analysis of variance with Satterthwaite's method. Differences among temperature and phosphorus treatments were compared (post-hoc test) separately after the linear model tests. GHG fluxes were modeled using generalized linear mixed-effects models (GLMMs) (glmmTMB function in the glmmTMB package), assuming gaussian error distributions (Brooks et al., 2017). Environmental factors were included as fixed predictors, while mesocosm identity and sampling month were treated as random intercepts. Mesocosm identity was included to account for repeated measurements from the same experimental unit, whereas sampling month was included to account for shared seasonal variation among observations collected during the same period. This random effect structure was selected a priori based on the experimental design and sampling schedule, with the aim of reducing pseudoreplication and non-independence among repeated measurements. All fixed predictors were standardized using z-score transformation (subtracting the mean and dividing by the standard deviation) to ensure comparability of parameter estimates across predictors (Gelman, 2008; Schielzeth, 2010; Harrison et al., 2018). The relative importance of each predictor was quantified as the ratio of the absolute value of its standardized fixed-effect coefficient (excluding the intercept) to the sum of the absolute values of all standardized fixed-effect coefficients. Spearman's correlation matrices were generated for environmental factors and treatment.

Piecewise structural equation models (piecewiseSEMs) (psem function in the piecewiseSEM package) were employed to quantitatively characterize the underlying cause-effect relationships among water quality, nutrients, macrophytes, phytoplankton, and greenhouse gas (GHG) fluxes for three different gases under constant warming, variable warming, and phosphorus loading treatments (Lefcheck, 2016). PiecewiseSEM estimates complex causal networks by fitting a series of linked linear models, enabling the evaluation of relationships among multiple ecosystem components. To represent multivariate ecosystem properties, observed variables were combined to construct composite variables (Lefcheck, 2016). Specifically, the composite variable water quality was derived from DO, pH, EC, DOC and Turb; the composite variable macrophytes was represented by the PVI and FPC; and the composite variable nutrients was constructed from TP, PO_4^{3-}, TN, NH_4^+, and NO_3^-. Weights were derived from linear models (lm function in the stats package) fitted prior to SEM construction. Phytoplankton was represented separately by Chl-*a*. These composite and observed variables were subsequently incorporated into the piecewiseSEM framework to evaluate their effects on GHG fluxes.

3. Results

3.1. Monthly and intergroup variation of environmental factors

Environmental variables and nutrient concentrations exhibited pronounced seasonality and showed distinct responses to warming and phosphorus enrichment. DO declined from January to August and increased from September to December (Fig. 1A), without significant effects of temperature, phosphorus, or their interaction ($p > 0.05$; Table 1). pH peaked in April and decreased thereafter (Fig. 1B); under phosphorus enrichment, extreme warming significantly increased pH, with significant main effects of temperature ($p = 0.0078$) and phosphorus ($p = 0.0074$), while their interaction remained non-significant ($p = 0.1876$; Table 1). EC followed a broadly similar temporal pattern across treatments but remained elevated under both warming treatments (Fig. 1C), driven by significant temperature and temperature–phosphorus interaction effects ($p < 0.05$; Table 1). DOC was consistently higher in the warming treatments (Fig. 1D), with temperature as the only significant driver ($p < 0.001$; Table 1). Chl-*a* also displayed clear seasonal fluctuations (Fig. 1E) and responded strongly to both warming and phosphorus enrichment ($p < 0.001$), with concentrations elevated under all treatments and a significant interaction term ($p < 0.05$; Table 1). Turbidity typically peaked from July to September and reached its highest values under constant warming combined with phosphorus enrichment (Fig. 1F); only the temperature effect was significant ($p < 0.05$; Table 1).

3.2. Monthly and intergroup variation in nutrient status and macrophyte structure

Nutrient dynamics exhibited seasonal peaks modulated by experimental treatments. TP exhibited similar seasonal patterns across treatments and was markedly elevated under constant warming with phosphorus enrichment (Fig. 2A), with temperature, phosphorus, and their interaction all exerting significant effects ($p < 0.01$; Table 2). PO_4^{3-} followed a comparable seasonal trajectory and was significantly influenced by temperature, phosphorus, and their interaction ($p < 0.05$; Fig. 2B; Table 2). TN fluctuated among treatments, reaching its maximum under constant warming with phosphorus enrichment and its minimum in the control during December (Fig. 2C); temperature was the only significant driver ($p < 0.001$; Table 2). NH_4^+ increased from January to October and subsequently declined (Fig. 2D), showing significant responses to temperature and phosphorus but not to their interaction (Table 2). NO_3^- decreased steadily from its January maximum to a minimum in May before rising again (Fig. 2E); phosphorus enrichment significantly reduced NO_3^- concentrations, and temperature, phosphorus, and their interaction all had significant effects ($p < 0.05$; Table 2). Aquatic macrophytes also exhibited clear seasonal dynamics. FPC increased initially and then declined across all treatments, peaking in July and September (Fig. 2F). Temperature was the only significant factor affecting FPC, with both constant and variable warming significantly reducing FPC ($p < 0.001$; Table 2). Additionally, the PVI showed a similar unimodal pattern but peaked earlier under warming treatments (April) than in the control (May; Fig. 2G). Phosphorus was the only significant determinant of PVI ($p < 0.05$; Table 2).

3.3. Monthly, intergroup and total annual variation of GHG fluxes

Environmental drivers produced clear seasonal cycles in greenhouse-gas fluxes and distinct, gas-specific responses to warming and phosphorus loading (Fig. 3; Table 3). Temperature exerted a significant main effect on CO_2 ($p = 0.0115$) and CH_4 fluxes ($p = 0.0174$), whereas phosphorus loading had a significant main effect only on N_2O ($p = 0.0156$); temperature × phosphorus interactions on CO_2, CH_4 and N_2O were not significant ($p > 0.05$) (Table 3).

CO_2 fluxes displayed a pronounced seasonal maximum in August

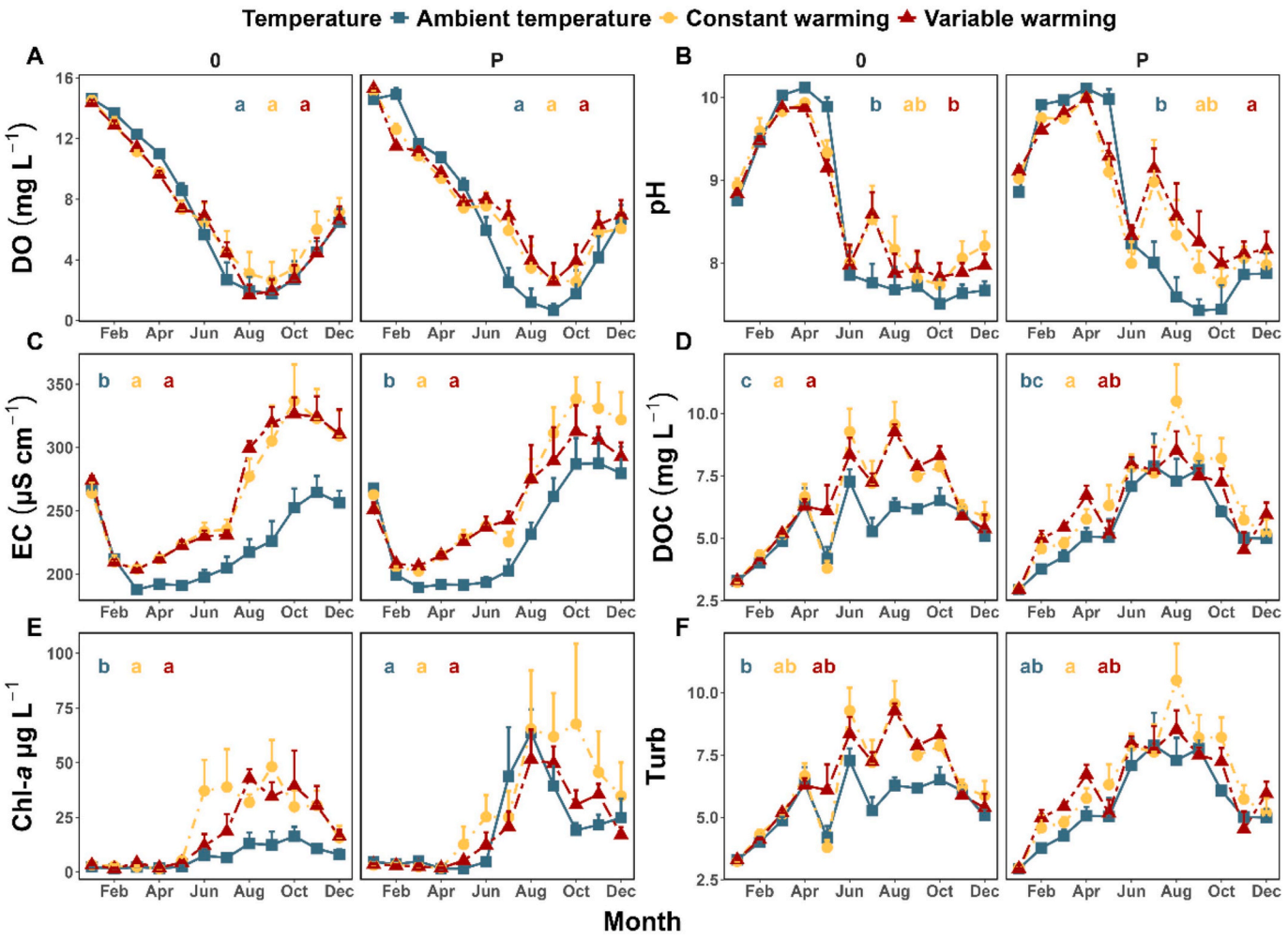


Fig. 1. Physico-chemistry (dissolved oxygen (DO), pH value (pH), electrical conductivity (EC), dissolved organic carbon (DOC) and turbidity (Turb)) and phytoplankton (chlorophyll-*a* (Chl-*a*)) (A-F) in the mesocosms under different treatments during the experimental period across different months. Means were compared for temperature treatments (ambient: blue, constant warming: yellow, variable warming: red) separately for each of the two phosphorus treatments (control: 0, phosphorus loading: P), with time as a co-variable. Different lowercase letters indicate significant differences between treatments. Vertical bars represent standard errors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Effects of temperature, phosphorus loading and their interactions on physico-chemistry and phytoplankton in the mesocosms. Logarithmic (log) or square root (sqrt) transformations were applied to meet the normality assumption and stabilize variance. Bold numbers indicate $p < 0.05$.

Response variable	Factors	Sum Sq	Mean Sq	NumDF	DenDF	*F*	*p*
DO	Temperature	12.96	6.48	2	415	1.49	0.2273
	Phosphorus	5.12	5.12	1	415	1.18	0.2790
	Temperature* Phosphorus	19.64	9.82	2	415	2.25	0.1064
pH	Temperature	2.61	1.30	2	415	4.91	**0.0078**
	Phosphorus	1.92	1.92	1	415	7.23	**0.0074**
	Temperature* Phosphorus	0.89	0.45	2	415	1.68	0.1876
EC	Temperature	108324.21	54162.11	2	415	56.80	**<0.0001**
	Phosphorus	106.89	106.89	1	415	0.11	0.7379
	Temperature* Phosphorus	5814.93	2907.46	2	415	3.05	**0.0485**
DOC	Temperature	71.25	35.63	2	415	17.71	**<0.0001**
	Phosphorus	0.01	0.01	1	415	0.00	0.9456
	Temperature* Phosphorus	2.76	1.38	2	415	0.69	0.5039
sqrt (Chl-*a*)	Temperature	80.62	40.31	2	415	13.37	**<0.0001**
	Phosphorus	42.31	42.31	1	415	14.04	**0.0002**
	Temperature* Phosphorus	22.59	11.30	2	415	3.75	**0.0244**
log (Turb)	Temperature	9.33	4.67	2	415	6.87	**0.0012**
	Phosphorus	1.70	1.70	1	415	2.51	0.1139
	Temperature* Phosphorus	0.18	0.09	2	415	0.13	0.8756

(Fig. 3A). Both constant and variable warming tended to attenuate the summer CO_2 peaks observed in ambient mesocosms, producing lower monthly emissions and reduced annual CO_2 totals relative to ambient conditions (Fig. 3A, 3D). Phosphorus enrichment likewise reduced

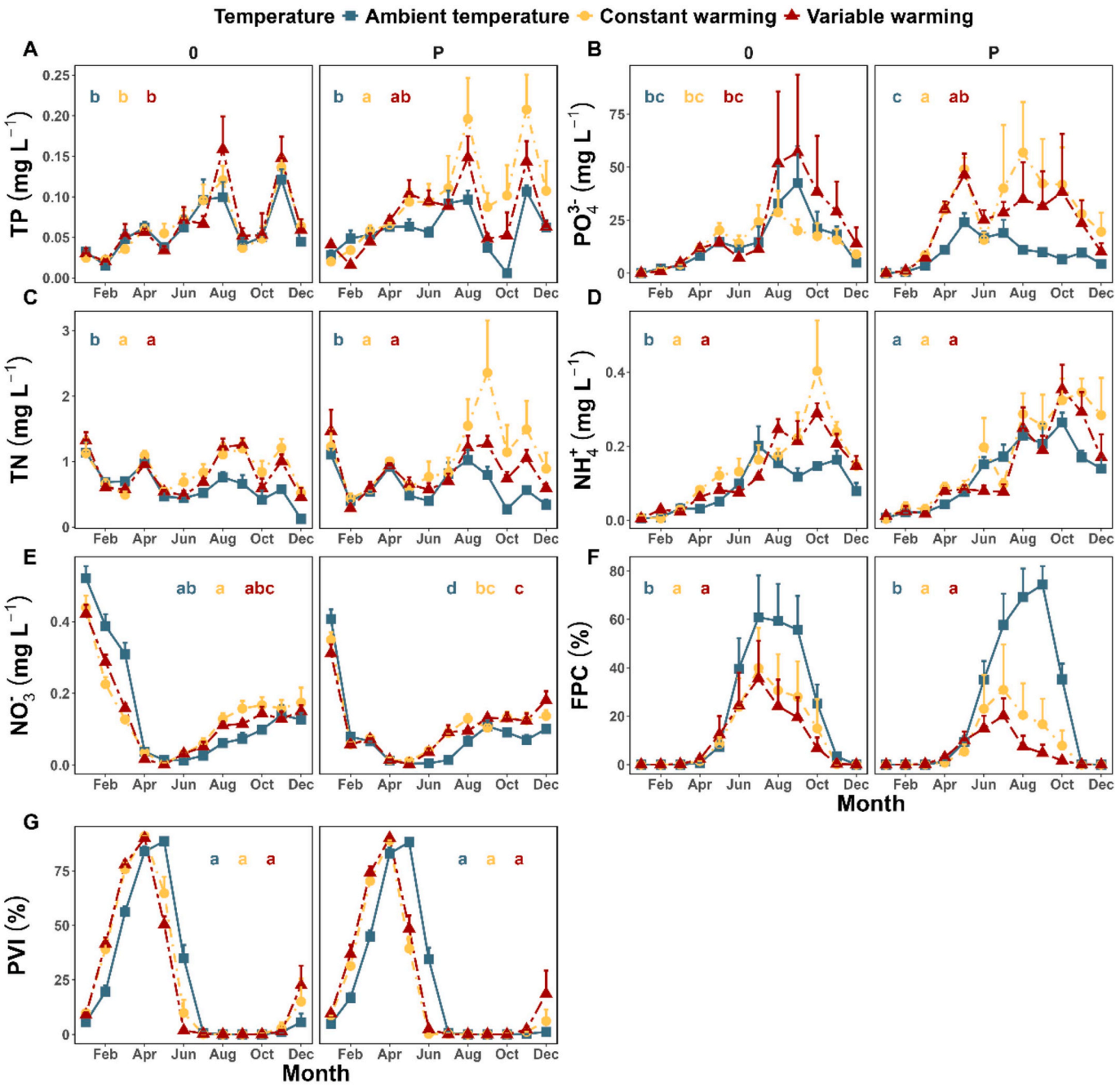


Fig. 2. Nutrients (total phosphorus (TP), phosphate (PO_4^{3-}), total nitrogen (TN), ammonium (NH_4^+) and nitrate (NO_3^-)) and macrophytes (submerged plant PVI (PVI) and floating-leaf plant coverage (FPC)) (A-G) in the mesocosms under different treatments during the experimental period across different months. Means were compared for temperature treatments (ambient: blue, constant warming: yellow, variable warming: red) separately for each of the two phosphorus treatments (control: 0, phosphorus loading: P), with time as a co-variable. Vertical bars represent standard errors. The treatment symbols are the same as given in Fig. 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

annual CO_2 emissions in our mesocosms (Fig. 3D). Additionally, CH_4 fluxes peaked earlier, in July (Fig. 3B), and responded differently: both warming regimes increased CH_4 emissions relative to ambient conditions, with the largest increases occurring when constant warming was combined with phosphorus enrichment (Fig. 3B, 3E). By contrast, N_2O fluxes were temporally variable and of much smaller magnitude than CO_2 or CH_4 (Fig. 3C). N_2O did not show a consistent response to temperature alone, but was significantly affected by phosphorus loading (reduced N_2O under P enrichment; $p = 0.0156$), and annual N_2O totals reflected this phosphorus effect (Fig. 3C, 3F). The data indicate that warming and phosphorus loading exert gas-specific controls, warming reduced CO_2 evasion while enhancing CH_4 emissions, whereas phosphorus loading depressed both annual CO_2 and N_2O emissions and amplified CH_4 responses when combined with constant warming.

3.4. *Effects of multiple environmental factors on GHG fluxes*

Greenhouse gas fluxes exhibited distinct but non-random responses to multiple environmental drivers (Fig. 4; Fig. S3), with a limited number of predictors consistently exerting strong effects across gases. For CO_2 fluxes, pH and FPC showed the largest standardized effect sizes and represented the most important predictors, together explaining a

Table 2
Effects of temperature, phosphorus loading and their interactions on nutrients and macrophytes in the mesocosms. Logarithmic (log) or square root (sqrt) transformations were applied to meet the normality assumption and stabilize variance. Bold numbers indicate $p < 0.05$.

Response variable	Factors	Sum Sq	Mean Sq	NumDF	DenDF	*F*	*p*
sqrt (TP)	Temperature	0.08	0.04	2	415	9.10	**0.0001**
	Phosphorus	0.06	0.06	1	415	13.75	**0.0002**
	Temperature* Phosphorus	0.04	0.02	2	415	4.76	**0.0091**
sqrt (PO_4^{3-})	Temperature	58.35	29.17	2	415	8.26	**0.0003**
	Phosphorus	22.77	22.77	1	415	6.44	**0.0115**
	Temperature* Phosphorus	47.22	23.61	2	415	6.68	**0.0014**
sqrt (TN)	Temperature	2.16	1.08	2	415	25.71	**<0.0001**
	Phosphorus	0.11	0.11	1	415	2.65	0.1043
	Temperature* Phosphorus	0.08	0.04	2	415	0.95	0.3861
sqrt (NH_4^+)	Temperature	0.21	0.10	2	415	10.31	**<0.0001**
	Phosphorus	0.09	0.09	1	415	8.52	**0.0037**
	Temperature* Phosphorus	0.04	0.02	2	415	1.86	0.1565
sqrt (NO_3^-)	Temperature	0.06	0.03	2	415	4.98	**0.0073**
	Phosphorus	0.33	0.33	1	415	53.49	**<0.0001**
	Temperature* Phosphorus	0.05	0.03	2	415	4.09	**0.0174**
PVI	Temperature	0.01	0.00	2	415	0.26	0.7745
	Phosphorus	0.08	0.08	1	415	6.02	**0.0146**
	Temperature* Phosphorus	0.03	0.02	2	415	1.33	0.2654
sqrt (FPC)	Temperature	2.18	1.09	2	415	22.35	**<0.0001**
	Phosphorus	0.05	0.05	1	415	0.97	0.3241
	Temperature* Phosphorus	0.16	0.08	2	415	1.62	0.1998

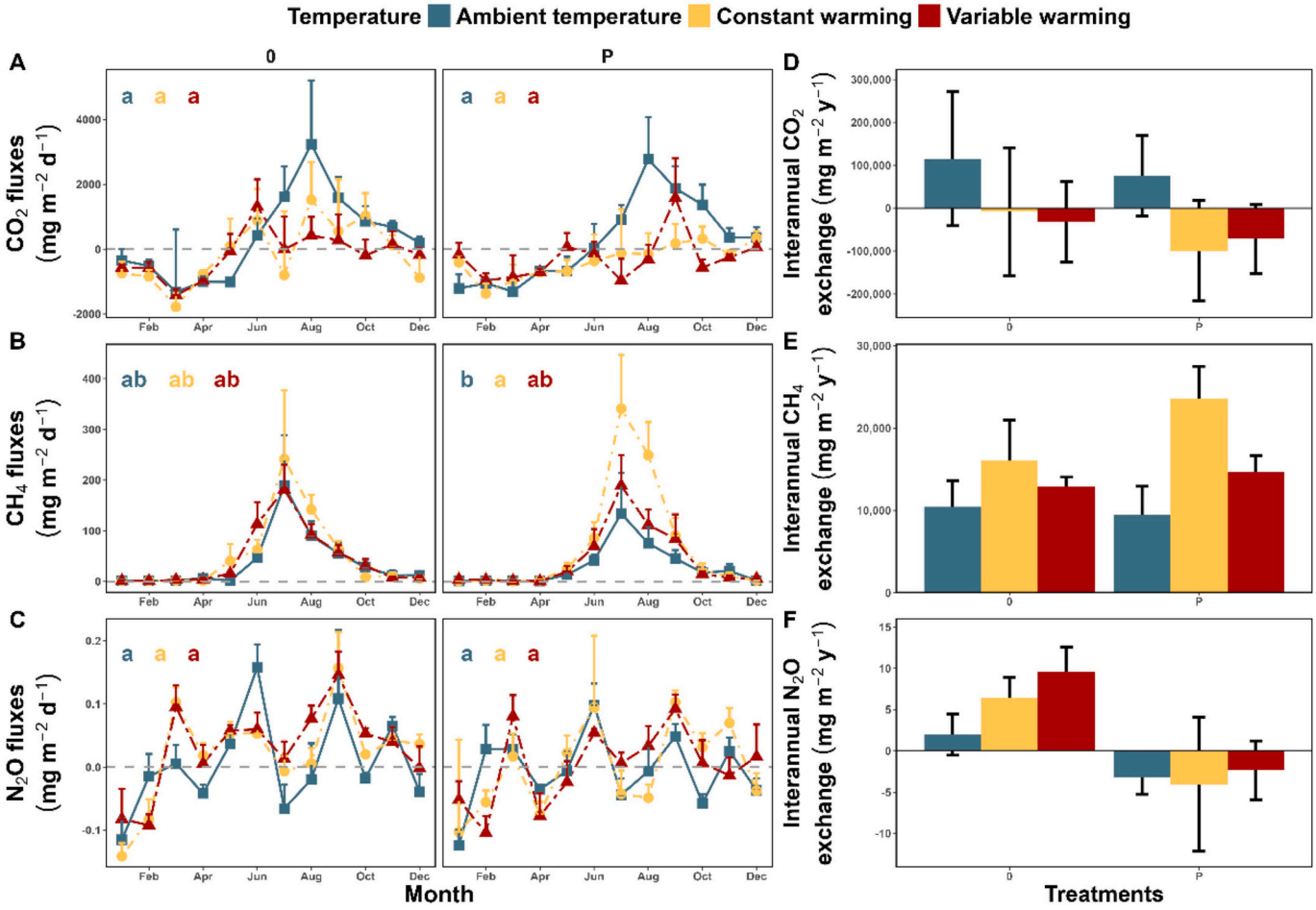


Fig. 3. GHGs (CO_2, CH_4, N_2O) fluxes in the mesocosms under different treatments during the experimental period across different months, with time as a co-variable (A-C). The total annual fluxes of CO_2, CH_4, or N_2O in the mesocosms under different treatments (D-F). Means were compared for temperature treatments (ambient: blue, constant warming: yellow, variable warming: red) separately for each of the two phosphorus treatments (control: 0, phosphorus loading: P). Vertical bars represent standard errors. The treatment symbols are the same as given in Fig. 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

substantial proportion of the total variance, followed by PVI, DO and DOC, whereas most nutrient variables contributed relatively little

Table 3
Effects of temperature, phosphorus loading and their interactions on GHGs (CO_2, CH_4, N_2O) fluxes in the mesocosms. Bold numbers indicate $p < 0.05$.

Response variable	Factors	Sum Sq	Mean Sq	NumDF	DenDF	*F*	*p*
CO_2 fluxes	Temperature	26166200.47	13083100.23	2	414.9996	4.51	**0.0115**
	Phosphorus	2543952.41	2543952.41	1	414.9996	0.88	0.3494
	Temperature* Phosphorus	162397.98	81198.99	2	414.9996	0.03	0.9724
CH_4 fluxes	Temperature	47845.70	23922.85	2	415	4.09	**0.0174**
	Phosphorus	1837.58	1837.58	1	415	0.31	0.5753
	Temperature* Phosphorus	15471.99	7735.99	2	415	1.32	0.2673
N_2O fluxes	Temperature	0.02	0.01	2	415	1.26	0.2850
	Phosphorus	0.05	0.05	1	415	5.90	**0.0156**
	Temperature* Phosphorus	0.01	0.00	2	415	0.34	0.7126

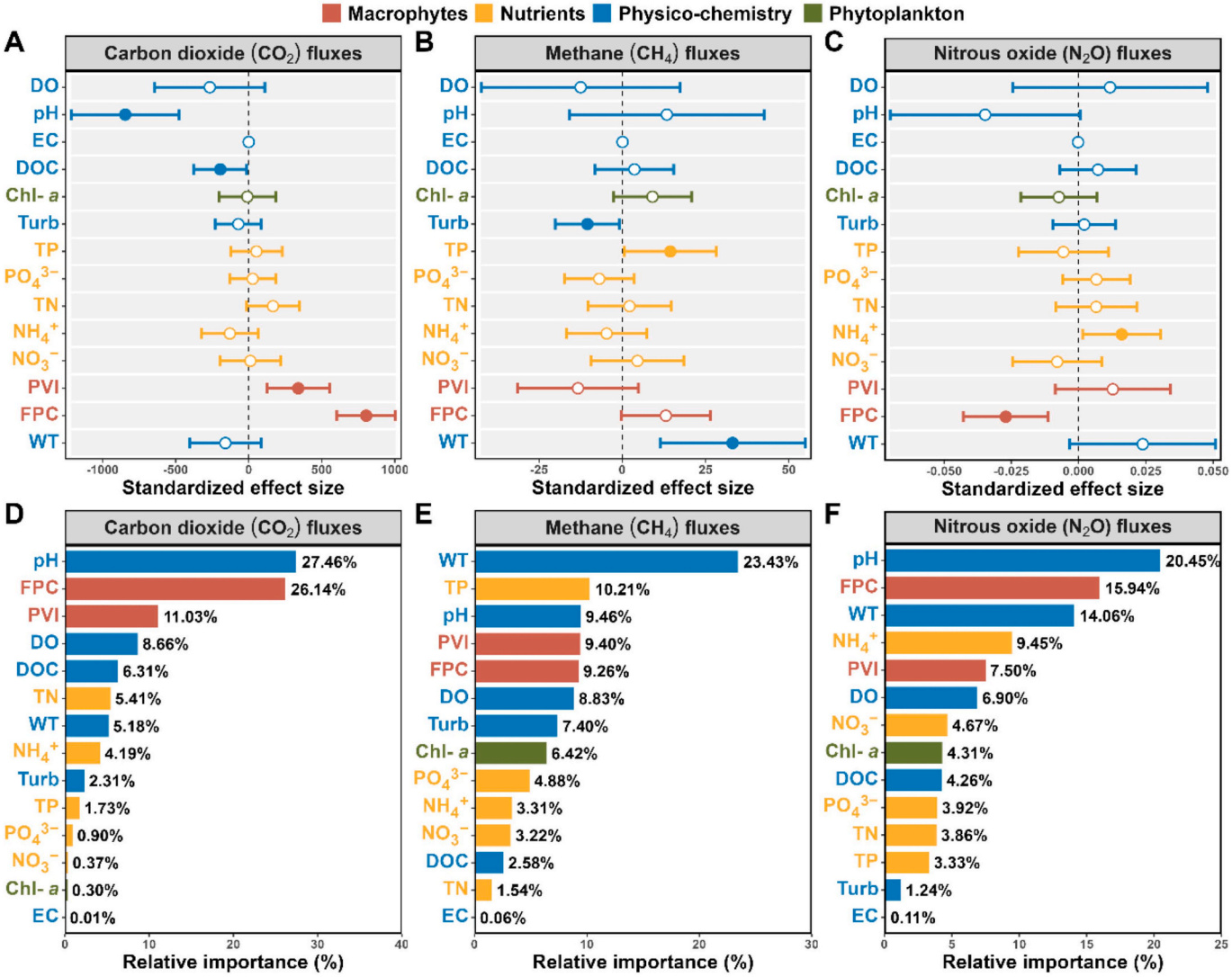


Fig. 4. Impacts of environmental drivers on GHG fluxes. Standardized effect sizes of predictor variables on CO_2, CH_4, and N_2O fluxes (A–C). Relative importance of predictor variables in explaining CO_2 fluxes, CH_4 fluxes and N_2O fluxes (D-F). The standardized effect size (±95% confidence intervals) was derived from GLMMs with a gaussian distribution, with predictor variables standardized by subtracting the mean and dividing by the standard deviation. Positive effects (the 95% confidence interval does not include zero) indicate significant positive relationships, while negative effects indicate the converse. The bar length represents the relative contribution of each predictor. Colors denote different categories of drivers (physico-chemistry, nutrients, macrophytes or phytoplankton).

(Fig. 4A, D). A similar pattern was observed for N_2O fluxes, for which pH and FPC were again the strongest predictors, with additional contributions from water temperature (WT), ammonium (NH_4^+) and PVI (Fig. 4C, F). In contrast, CH_4 fluxes were more strongly influenced by physico-chemical and macrophyte-related factors, with WT, TP, pH, PVI, and FPC emerging as the dominant drivers (Fig. 4B, E). Marginal effect analyses further showed that these key drivers displayed consistent directional responses across their observed gradients, with positive or negative slopes depending on gas type, while predictors with low relative importance exhibited weak or near-flat responses (Fig. S3). These results indicate that CO_2, CH_4, and N_2O fluxes are controlled by distinct biogeochemical and ecological drivers.

3.5. Environmental regulation of GHG fluxes under treatments

Piecewise structural equation modeling integrated warming treatments and phosphorus loading with changes in physico-chemistry, nutrient status, and producer composition to examine how environmental drivers were associated with greenhouse-gas fluxes within a unified system framework (Fig. 5A-C). For CO_2 fluxes (Fig. 5A), physico-chemistry showed the strongest direct positive association with CO_2 (SPC = 0.419, $p < 0.001$), followed by a positive effect of macrophytes (SPC = 0.360, $p < 0.001$). In contrast, neither nutrients nor phytoplankton showed significant relationships with CO_2. Both constant and variable warming were negatively associated with physico-chemistry (SPC = − 0.243 and − 0.240, respectively; $p < 0.001$) and with macrophytes (SPC = − 0.200 and − 0.249, respectively; $p < 0.001$), while phosphorus loading was positively associated with phytoplankton (SPC = 0.139, $p < 0.001$). Neither warming nor phosphorus loading exerted a significant direct effect on CO_2 fluxes, indicating predominantly indirect regulation pathways. For CH_4 fluxes (Fig. 5B), macrophytes exhibited the strongest positive association with CH_4 emissions (SPC = 0.286, $p < 0.001$), followed by physico-chemistry (SPC = 0.216, $p < 0.01$). The effects of nutrients and phytoplankton on CH_4 were not significant. In addition, constant and variable warming were positively associated with nutrients (SPC = 0.194, $p < 0.001$ and SPC = 0.107, $p < 0.05$, respectively) and negatively associated with macrophytes (SPC = − 0.162 and − 0.206, $p < 0.01$, respectively). Phosphorus loading was positively associated with nutrients (SPC = 0.204, $p < 0.001$) and macrophytes (SPC = 0.139, $p < 0.001$). In contrast to CO_2, constant (SPC = 0.159, $p < 0.001$) and variable (SPC = 0.104, $p < 0.001$) warming exerted significant direct positive effects on CH_4 fluxes. For N_2O fluxes (Fig. 5C), nutrient status was the dominant positive driver (SPC = 0.279, $p < 0.01$), whereas associations with physico-chemistry, macrophytes and phytoplankton were weak and not significant. Constant warming (SPC = 0.158, $p < 0.01$), variable warming (SPC = 0.112, $p < 0.05$) and phosphorus loading (SPC = 0.193, $p < 0.001$) were positively associated with nutrients. In addition, constant warming (SPC = 0.198, $p < 0.001$) and phosphorus loading (SPC = 0.139, $p < 0.01$) were positively associated with phytoplankton. Notably, phosphorus loading exhibited a significant direct negative association with N_2O fluxes (SPC = − 0.154, $p < 0.01$).

4. Discussion

4.1. Effects of warming and phosphorus enrichment on CO_2 emissions

The principal finding of this study is that experimental warming reduced annual CO_2 emissions from these macrophyte-dominated mesocosms. The statistical relationships identified in this study suggest that warming did not substantially influence CO_2 emissions through direct thermal effects. Instead, its influence was primarily associated with indirect changes in macrophyte community structure and water chemistry. Specifically, warming promoted submerged macrophyte growth and altered water physicochemical conditions by increasing pH and dissolved oxygen. Enhanced photosynthetic carbon uptake by submerged macrophytes likely reinforced these physicochemical changes (Wang et al., 2025). Higher pH can shift carbonate equilibrium away from free CO_2, thereby reducing the dissolved CO_2 pool available for emission. Meanwhile, the decline in floating-leaved plant cover may have altered underwater light conditions and air–water gas exchange, because floating-leaved canopies can act as a partial physical barrier at the water surface (Ribaudo et al., 2012). These mechanisms suggest that warming reduced CO_2 emissions primarily through macrophyte-mediated ecosystem restructuring rather than through direct thermal stimulation of carbon mineralization.

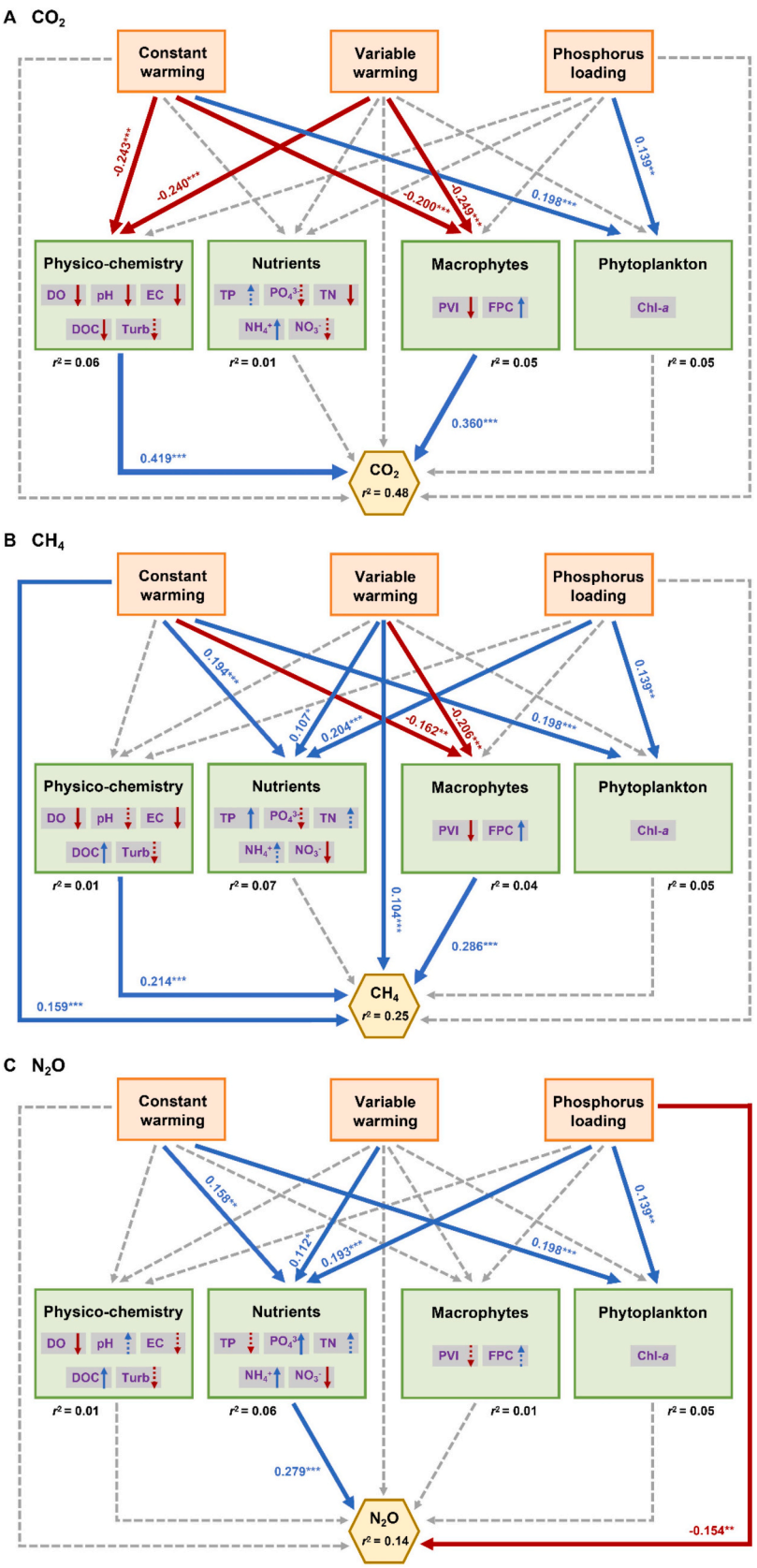
These findings differ from the commonly reported positive temperature dependence of aquatic CO_2 production and release (Marotta et al., 2014; Yvon-Durocher et al., 2010b), but are consistent with increasing evidence that warming effects on inland-water carbon cycling are strongly ecosystem dependent (Yuan et al., 2024; Yvon-Durocher et al., 2010a). In this study, CO_2 dynamics were more closely associated with temporal temperature variation than with the warming treatments themselves, suggesting that the imposed 3.5°C warming may not have exceeded the threshold required to stimulate substantial CO_2 release at the ecosystem scale. The similar responses under constant and variable warming further imply that ecosystem state, including nutrient conditions and biotic interactions, constrains the expression of warming effects. This interpretation is supported by previous mesocosm studies showing that heat-wave events may have negligible effects on CO_2 emissions under certain nutrient regimes because biological interactions can override direct temperature effects (Audet et al., 2017). Collectively, these results indicate that ecosystem structure, rather than warming intensity alone, may determine the direction and magnitude of warming-induced changes in CO_2 emissions from shallow lakes.

Phosphorus addition further reduced CO_2 emissions under warming, although its overall direct effect was weak. Its influence was mainly associated with phytoplankton-related pathways; however, phytoplankton and dissolved nutrient pools contributed relatively little to CO_2 production in this macrophyte-rich system. This contrasts with findings from some eutrophic lakes, where nutrient availability and phytoplankton biomass have been reported to exert stronger control over CO_2 dynamics (Kosten et al., 2012; Pacheco et al., 2014). These contrasting responses suggest that vegetation structure may mediate how nutrient enrichment influences carbon cycling and can modulate, or even outweigh, the direct effects of warming in macrophyte-rich shallow lakes. Nevertheless, the macrophyte abundance or coverage threshold at which this vegetation control emerges remains unresolved and should be addressed in future studies.

4.2. Effects of warming and phosphorus enrichment on CH_4 emissions

In this study, the CH4 emissions responded fundamentally differently from CO_2, being governed primarily by the direct effects of warming rather than by indirect ecosystem feedbacks. Under both constant and variable warming, CH_4 fluxes increased (Fig. 3), consistent with the strong temperature dependence of methanogenesis. Methanogenic activity typically increases exponentially within ecologically relevant thermal ranges, thereby enhancing CH_4 production potential (Yang et al., 2024), and warming can disproportionately stimulate ebullitive CH_4 release from freshwaters, with emissions increasing by approximately 6–20% per °C (Aben et al., 2017). The observed relationships support direct thermal stimulation as the dominant mechanism regulating CH_4 production and emission.

These findings are consistent with previous studies demonstrating that CH_4 production is generally more temperature-sensitive than CO_2 production in freshwater ecosystems, but they also indicate that the magnitude of the warming response depends on ecosystem structure (Yuan et al., 2024). Although warming consistently enhanced CH_4 emissions, warming-induced shifts in macrophyte community composition created an opposing indirect pathway that partially offset this stimulation. Warming increased submerged macrophyte abundance while reducing floating-leaved plant cover, and this shift in macrophyte community composition toward submerged macrophytes has been shown to reduce CH_4 emissions from shallow lakes by increasing the temperature threshold for substantial ebullition (Aben et al., 2022). Accordingly, the increase in CH_4 emissions under variable warming was weaker than under constant warming, likely because variable warming promoted a greater increase in submerged macrophytes together with a stronger decline in floating-leaved vegetation (Fig. 5B). Studies in similar mesocosm systems have likewise reported differences between constant and variable warming in their effects on submerged macrophyte reproduction (Xu et al., 2020). These results suggest that while warming provides the primary thermal driver of CH_4 production, vegetation-mediated ecosystem feedbacks can substantially regulate the



(caption on next page)

Fig. 5. Piecewise structural equation modeling (PSEM) shows that constant warming, variable warming, and phosphorus loading influence CO_2, CH_4, and N_2O fluxes through physico-chemistry, nutrients, macrophytes, and phytoplankton. Physico-chemistry is a composite variable constructed from DO, pH, EC, DOC, and Turb; nutrients from TP, PO_4^{3-}, TN, NH_4^+, and NO_3^-; macrophytes from PVI and FPC; and phytoplankton from Chl-*a*. Arrows in boxes indicate correlations between indicators and composite variables. Boxes denote variables; arrows denote relationships. Blue and red arrows indicate significant positive and negative relationships, respectively; grey dashed arrows indicate non-significant effects. Numbers next to arrows are standardized path coefficients, "*" indicates significance (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), and r^2 values for endogenous variables are shown next to boxes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

magnitude of the emission response.

Phosphorus addition had no significant direct effect on CH_4 emissions, but it enhanced warming-induced CH_4 release (Fig. 3), likely by stimulating algal production and increasing the supply of labile algal-derived organic matter available to methanogens (Davidson et al., 2018; Niu et al., 2024). This observation further supports the view that phosphorus enrichment alone may be insufficient to alter CH_4 emissions in macrophyte-dominated shallow lakes, whereas its effects become more evident when interacting with climate warming. Overall, these findings indicate that CH_4 emissions from shallow lakes are primarily constrained by thermal conditions, whereas the magnitude of their response is modulated by vegetation structure and nutrient-mediated changes in substrate availability.

4.3. Effects of warming and phosphorus enrichment on N_2O emissions

N_2O emissions were more strongly regulated by phosphorus enrichment than by warming. Although warming tended to increase N_2O fluxes, the effect was not statistically significant, whereas phosphorus addition shifted the system from an annual net source to an annual net sink of N_2O (Fig. 3; Table 3). Constant and variable warming produced broadly similar responses, suggesting that the net warming effect on N_2O exchange reflected a balance between opposing processes. Warming can enhance internal nutrient release, stimulate phytoplankton growth, and increase oxygen consumption, thereby creating microsites favorable for nitrification and denitrification-derived N_2O production, particularly when organic carbon availability is high (Audet et al., 2017; Chen et al., 2024; Kosten et al., 2009). At the same time, enhanced primary production may also reduce dissolved inorganic nitrogen availability, thereby limiting substrate supply for N_2O formation. These opposing mechanisms likely explain the weak net response of N_2O emissions to warming observed in this study.

By contrast, phosphorus exerted a stronger, although more complex, influence on N_2O dynamics. Phosphorus addition increased NH_4^+ and decreased NO_3^- concentrations (Fig. 5C), potentially creating conditions favorable for microbial pathways associated with N_2O production (Velthuis and Veraart, 2022; Veraart et al., 2011). However, phosphorus also had a significant direct negative effect on N_2O fluxes, indicating that its suppressive influence dominated at the ecosystem scale. One plausible explanation is that elevated phosphorus availability promotes microbial nitrogen transformation and enhances the completeness of denitrification, thereby reducing the accumulation of the intermediate product N_2O (Liang et al., 2025). Phosphorus enrichment therefore regulated not only the magnitude of N_2O exchange, but also its direction, highlighting the importance of nutrient-mediated shifts in nitrogen cycling pathways for greenhouse-gas production.

Overall, the three greenhouse gases exhibited fundamentally different regulatory pathways in response to warming and phosphorus enrichment. CO_2 responses were dominated by indirect ecological feedbacks associated with macrophyte restructuring, whereas CH_4 responses were governed primarily by direct thermal stimulation but were moderated by shifts in macrophyte community composition. In contrast, N_2O dynamics were more strongly constrained by phosphorus-mediated nitrogen transformations. Together, these findings highlight macrophyte community structure as a common ecological regulator of carbon-related greenhouse-gas dynamics, although its influence differs between CO_2 and CH_4, whereas phosphorus availability exerts stronger control over N_2O exchange. These contrasting controls demonstrate that greenhouse-gas responses to global change cannot be generalized across gases because each gas is regulated by distinct combinations of thermal forcing, nutrient availability, and vegetation-mediated ecosystem feedbacks. Incorporating these gas-specific regulatory pathways into predictive models will improve projections of greenhouse-gas emissions from shallow lakes under future climate warming and eutrophication.

5. Conclusion

This study demonstrates that climate warming and phosphorus enrichment regulate greenhouse gas emissions in shallow lakes through distinct and gas-specific pathways, leading to differentiated environmental risks under multiple stressors. Carbon-related fluxes were primarily shaped by warming-induced ecosystem restructuring, especially shifts in macrophyte functional composition and associated water-chemistry changes, whereas N_2O dynamics were governed mainly by phosphorus-driven nitrogen transformations. The weaker CH_4 response under variable warming than under constant warming further indicates that warming regime can modulate methane emission risk beyond the effect of mean temperature alone. These findings highlight that nutrient enrichment and climate warming do not generate uniform responses of shallow-lake greenhouse gas emissions, and that single-gas or single-driver assessments may underestimate ecosystem responses under future global change. Future research should determine whether these regulatory pathways vary among lake types, particularly between eutrophic and oligotrophic systems, and identify the environmental thresholds at which warming and phosphorus enrichment interact synergistically or antagonistically. Such efforts will improve process-based predictions of greenhouse gas emissions from shallow lakes and enhance assessments of their climate feedbacks under future environmental change.

CRediT authorship contribution statement

Xiaoqi Xu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bowen Hu:** Writing – review & editing, Validation. **Danni Yuan:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization. **Guohuan Su:** Validation, Investigation, Data curation. **Lan Wang:** Validation, Investigation, Data curation. **Ziqi Chen:** Validation, Investigation. **Kangshun Zhao:** Validation, Data curation. **Tao Wang:** Validation, Data curation. **Jun Xu:** Validation, Funding acquisition, Data curation. **Min Zhang:** Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2026.136152.

Data availability

Data will be made available on request.

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